

Predictive Maintenance Model - Point-wise Summary

1. Project Purpose

- Predictive maintenance for equipment health monitoring.
- Current implementation is focused on motor scenario.

2. Programming Language Used

- Backend/ML: Python
- Frontend/UI: JavaScript (React)

3. ML Model Used

- Model: Logistic Regression (scikit-learn)
- Type: Binary classification (Failure vs No Failure)

4. Input Parameters (Time-Series)

- Torque
- Pressure
- Temperature
- Minimum 10 points required per parameter.

5. Data Used

- Dummy/simulated correlated time-series data for training/demo.
- Training sample generation includes trends, noise, and drift.

6. Feature Engineering

- Mean, standard deviation, slope (start vs end), and parameter correlations are extracted.

7. Correlation Handling

- Correlation matrix is computed among torque, pressure, temperature.
- Exposed through API summary endpoint.

8. API Endpoints

- GET /predictive-maintenance/summary
- GET /predictive-maintenance/sample-data
- POST /predictive-maintenance/predict
- POST /predict (now supports both generic and time-series format)

9. Output from Prediction

- predictedFailure (true/false)
- failureProbability (0 to 1)
- healthScore (0 to 100)
- recommendedAction (maintenance guidance)

10. Frontend Integration

- AI dashboard consumes summary + prediction APIs.

- Search-based detail card with auto-scroll and trend visibility.

11. Current Scope and Limitation

- Production model currently tuned for motor-type behavior.
- For other equipment, dedicated parameter mapping/training is needed.

12. Deployment/Run Basics

- Backend: `uvicorn api:app --reload --host 127.0.0.1 --port 8000`
- Frontend: `npm run dev`
- API docs: <http://127.0.0.1:8000/docs>

Prepared for external technical discussion.